H.T No: | | | | | | | | | | | **R18** Course Code: A30461

CMR COLLEGE OF ENGINEERING & TECHNOLOGY

(UGC AUTONOMOUS)

B.Tech III Semester Regular & Supplementary Examinations Feb/March-2023

Course Name: ANALOG & DIGITAL ELECTRONICS

(Common for CSE & IT)

Date: 22.02.2023 AN **Time: 3 hours** **Max.Marks: 70**

(Note: Assume suitable data if necessary)

PART-A

Answer all TEN questions (Compulsory)

Each question carries TWO marks. **10x2=20M**

1. Draw the Diode Equivalent Circuits. 2 M
2. What are the applications of Zener diode. 2 M
3. Compare Common Base, Common Emitter and Common Collector Amplifier Configurations in terms of AV, AI, Ri and R0. 2 M
4. Classify the types of transistors with its symbols. 2 M
5. Sketch the basic construction of a p-channel JFET. 2 M
6. Solve the following 2 M
   i) Convert (101011)2 into Gray code.
   ii) Convert the Gray code (101101) into a binary number.
7. Draw the logic circuit for single-bit magnitude comparator. 2 M
8. Implement the Boolean function F= AB + CDE + F using NAND gates only. 2 M
9. Compare the combinatorial circuits over sequential circuits? 2 M
10. Write excitation table for JK flip-flop. 2 M

PART-B

Answer the following. Each question carries TEN Marks. **5x10=50M**

11.A). Explain the operation of P-N junction diode under forward and reverse bias with necessary diagrams. 10M

OR

11. B). Draw the circuit diagram of Full wave rectifier? Explain the operation of circuit with relevant waveforms. 10M

12. A). Sketch the common emitter configuration circuit and explain the Input and output characteristics with neat diagrams. 10M

OR

12. B). Explain the construction, principle of operation, characteristics and applications of UJT. 10M

13. A). Draw and Explain about the Operation of N-Channel Field Effect Transistor. 10M

OR

13. B). i) Evaluate signed 10s complement form and perform the following operations: (9286) + (-801). 4M

ii) Generate the 12 bit Hamming code, for given the 8-bit data word 10101111 that detects & corrects single bit error if any. 6M

(P.T.O..)

14. A). Realize the Function F (A, B, C, D) = π M (0,2,3,5,6,8,10,12,14) using K-map and design logic diagram using Basic Gates. 10M

OR

14. B). Define Demultiplexer and explain the procedure to implement 16X1 Multiplexers by using 4X1 Multiplexers. 10M

15. A). i) Develop the Mealy machine state diagram from the given state table 7M

PS	NS,Z	
	X=0	X=1
A	C,0	B,0
B	A,1	D,0
C	B,1	A,1
D	D,1	C,0

ii) What are the capabilities and limitations of finite state machines? 3M

OR

15. B). Draw the circuit diagram of 4 bit ring counter using D flip-flops and explain its operation with the help of bit pattern. 10M

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